Internet of Things: Protocols and Networks

(CSC2106)

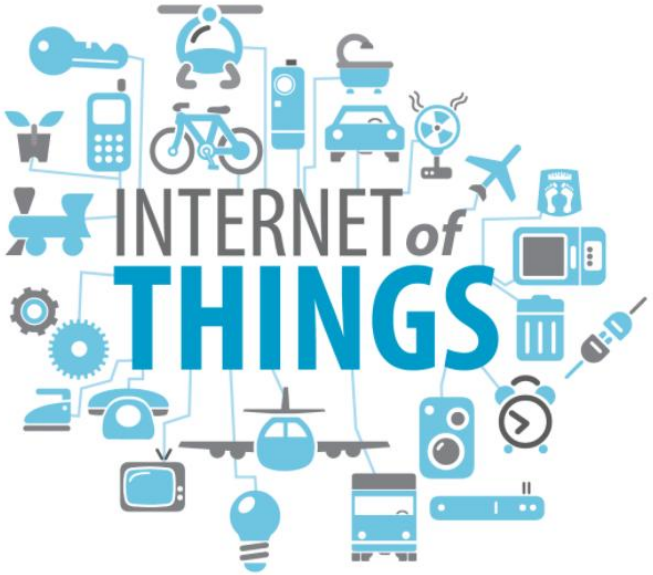
# Introduction to IoT

# So, what is IoT?

- The Internet of things (IoT) describes the network of physical objects—"things"—that are embedded with sensors, software, and other technologies for the purpose of connecting and exchanging data with other devices and systems over the Internet. **(Wikipedia)**

- The Internet of Things (IoT) describes the network of physical objects—"things"—that are embedded with sensors, software, and other technologies for the purpose of connecting and exchanging data with other devices and systems over the internet. These devices range from ordinary household objects to sophisticated industrial tools. With more than 7 billion connected IoT devices today, experts are expecting this number to grow to 10 billion by 2020 and 22 billion by 2025. **(Oracle)**

- The Internet of Things has become a central and exciting research area encompassing many fields in information and communication technologies and adjacent domains. IoT systems involve interactions with heterogeneous, distributed, and intelligent things, both from the digital and physical worlds including the human in the loop. Thanks to the increasingly wide spectrum of applications and cheap availability of both network connectivity and devices, a number of different stakeholders from industry, academia, society and government are part of the IoT ecosystem. **(IoT Conference 2020)**

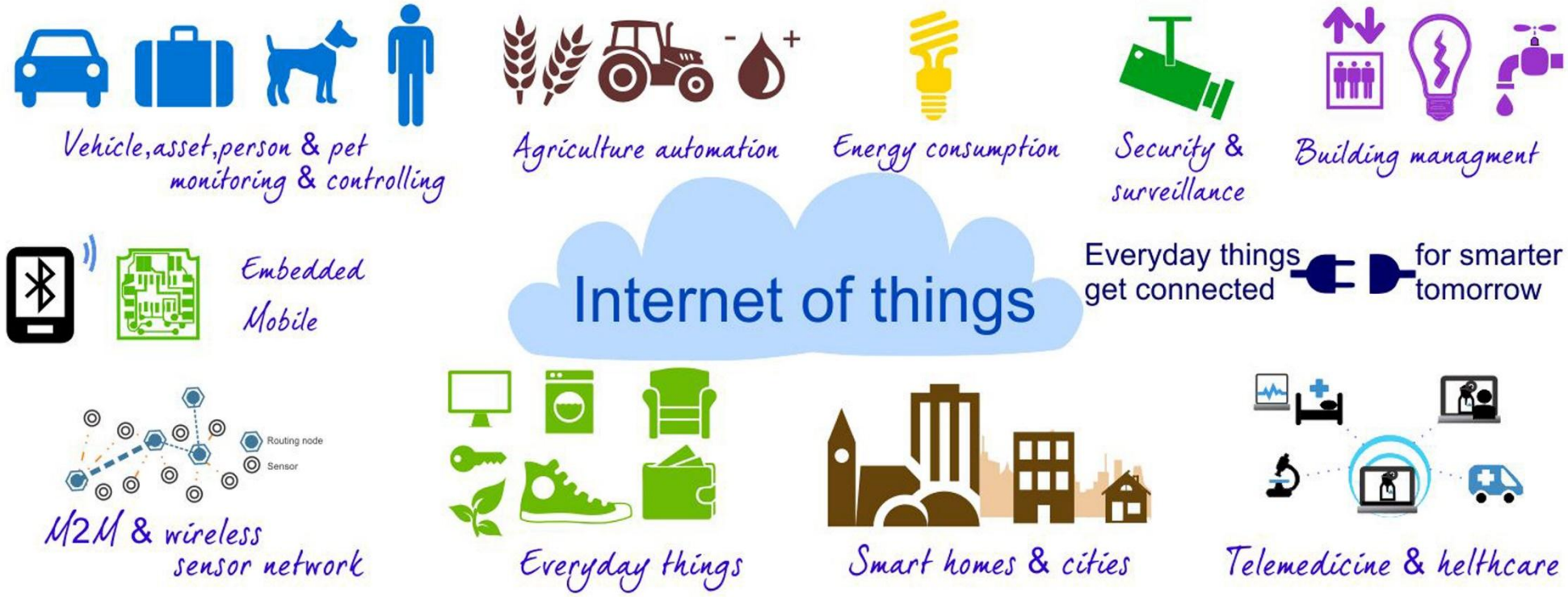


# IEEE definition of IoT

"An IoT system is a network of networks where, typically, a massive number of objects, things, sensors or devices are **connected** through communications and information infrastructure to provide value-added services via **intelligent** data processing and management for different **applications** (e.g. smart cities, smart health, smart grid, smart home, smart transportation, and smart shopping)."

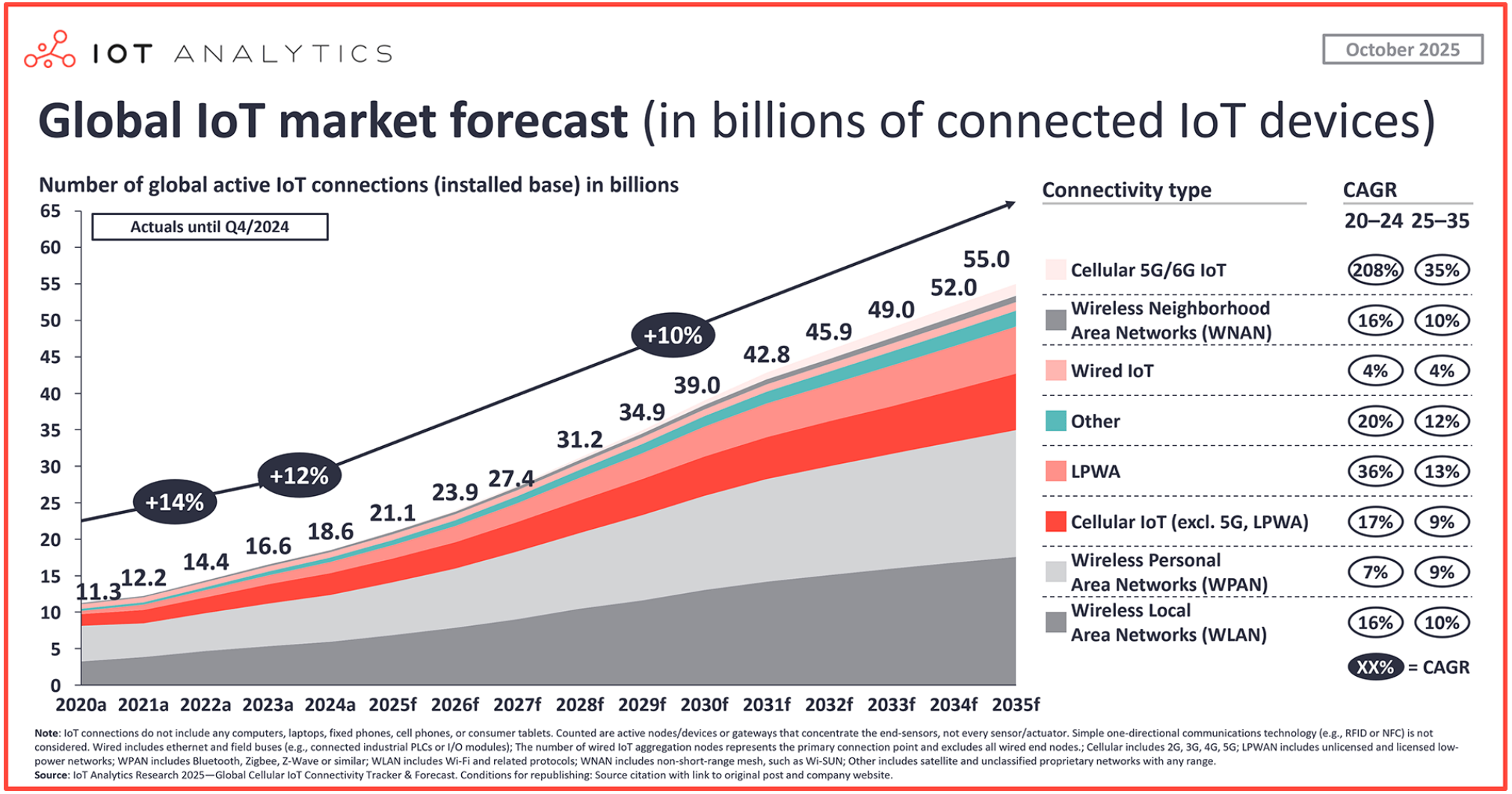
(IEEE Internet of Things Journal)

# ‘Thing’ in Internet-of-Things (IoT)




[https://inventrom.files.wordpress.com/2014/11/things.jpg]

# “Things” connected to the internet




Source: https://iot-analytics.com/number-connected-iot-devices/

# Internet-of-Things (IoT): Components

**Thing**
**(Physical object)**

=

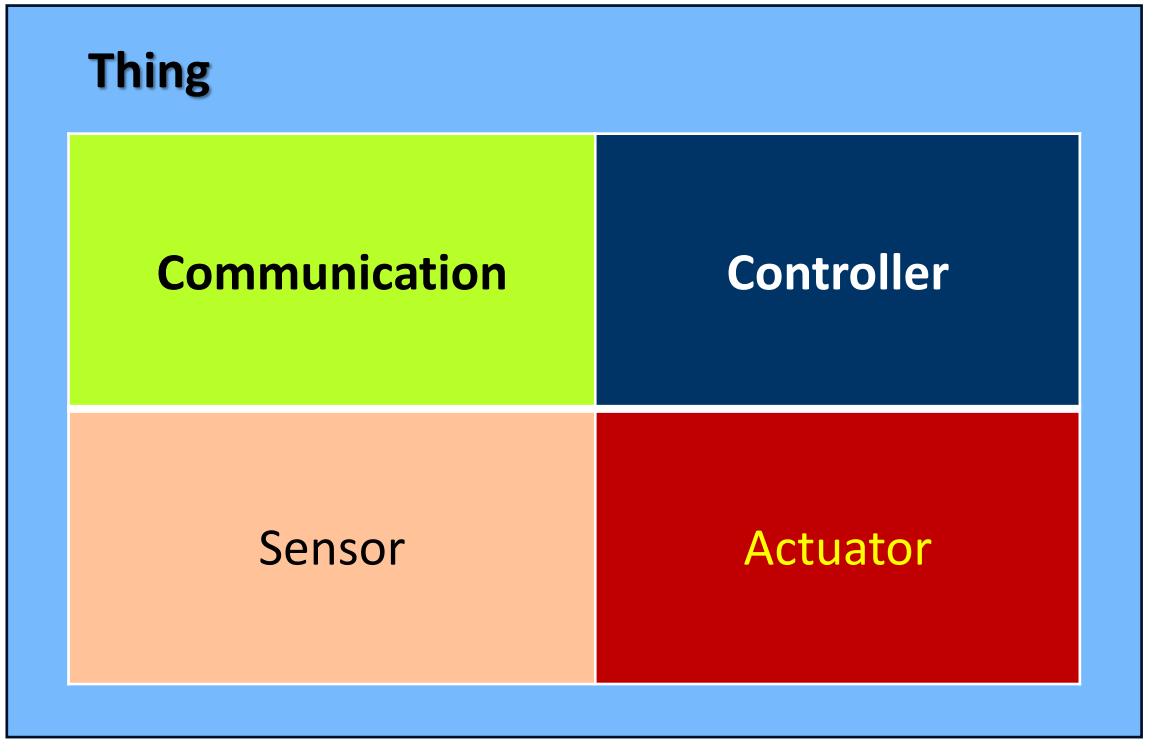
Controller ("Brain")

+

Sensors / Actuators

+

Networks (Internet)



# Internet-of-Things (IoT): Purpose

| COLLECT | CONNECT | COMPREHEND |
| --- | --- | --- |

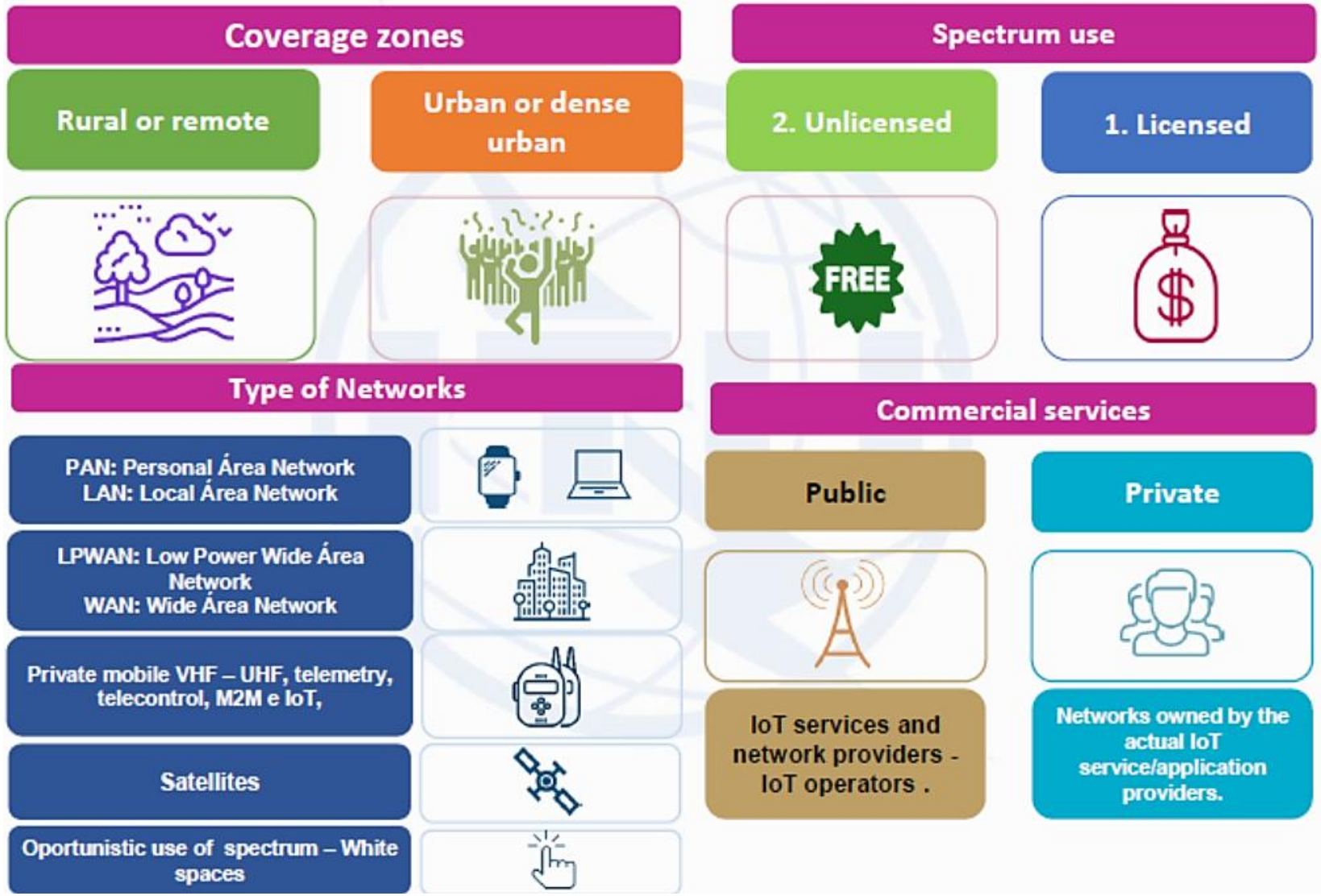
# Data vs. Knowledge

*"The ultimate goal is transforming the raw IoT data into actionable insights for decision-making and automation, leveraging real-time analytics and visualization."*

*This usually requires data from multiple sources, (near) real-time analytics and visualisation and/or semantic representations.*

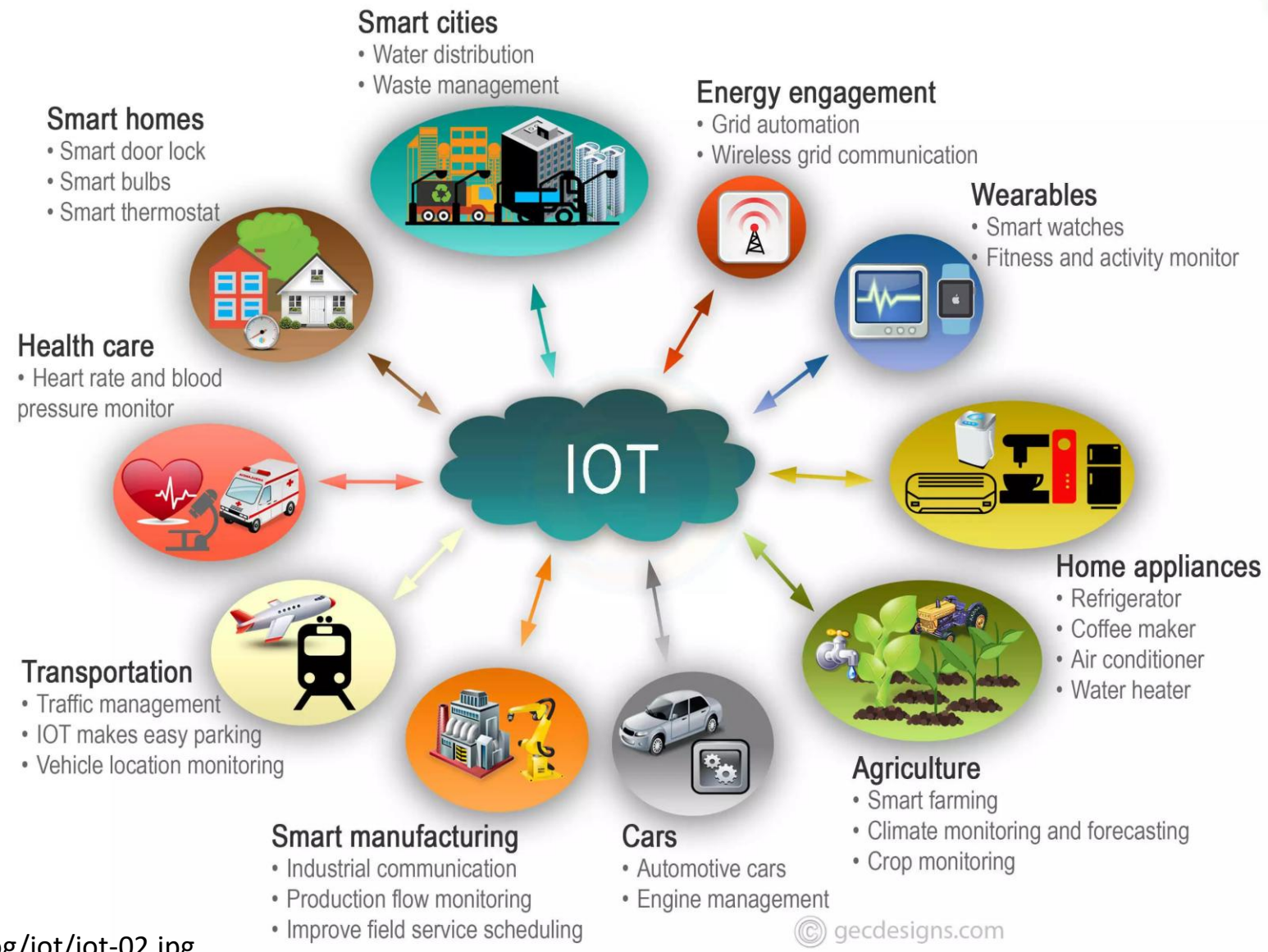
N. Moghe, S. Agarwal, V. Patel, A. Khandge and M. Nemade, "Exploring The Integration of Data Analytics and Internet of Things (IoT) for Smart Decision-Making In Various Industries," 2023 6th International Conference on Advances in Science and Technology (ICAST), Mumbai, India, 2023, pp. 40-45.

# IoT is Pervasive




*Source: ITU Workshop on Spectrum Management for Internet of Things Deployment, 22 November 2016, Geneva*

# IoT Applications Spectrum

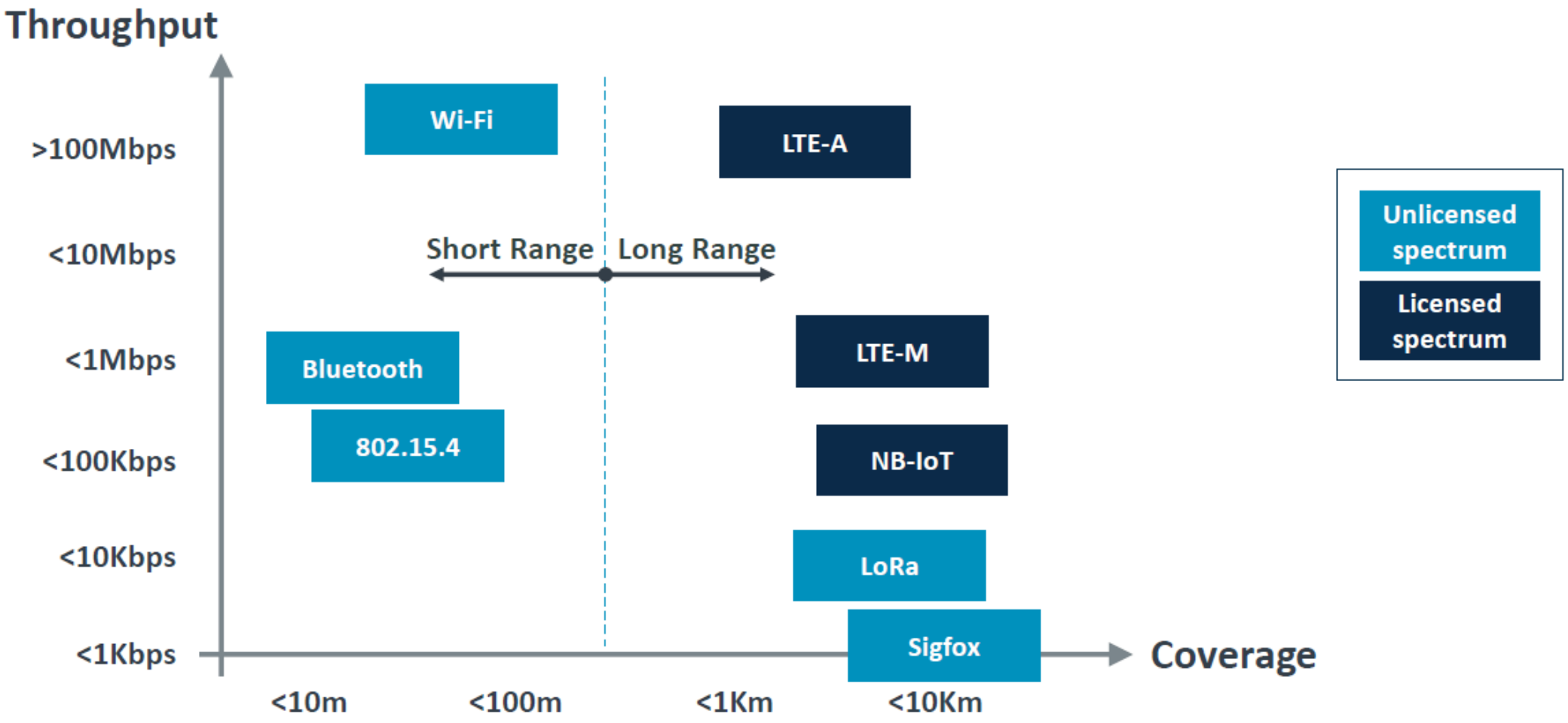



https://gecdesigns.com/img/blog/iot/iot-02.jpg

# Characteristics of IoT Communications

- **Wireless**
- Low cost
- Low power (long battery duration)
- Low processing capacity
- Low storage capacity
- Extensive network of connections (scalability)
- Various bitrate/bandwidth requirements
- Various range
- Various QoS requirements
- Small size devices (typically)
- Simple network architecture and protocols (arguably)

# IoT Connectivity Technologies

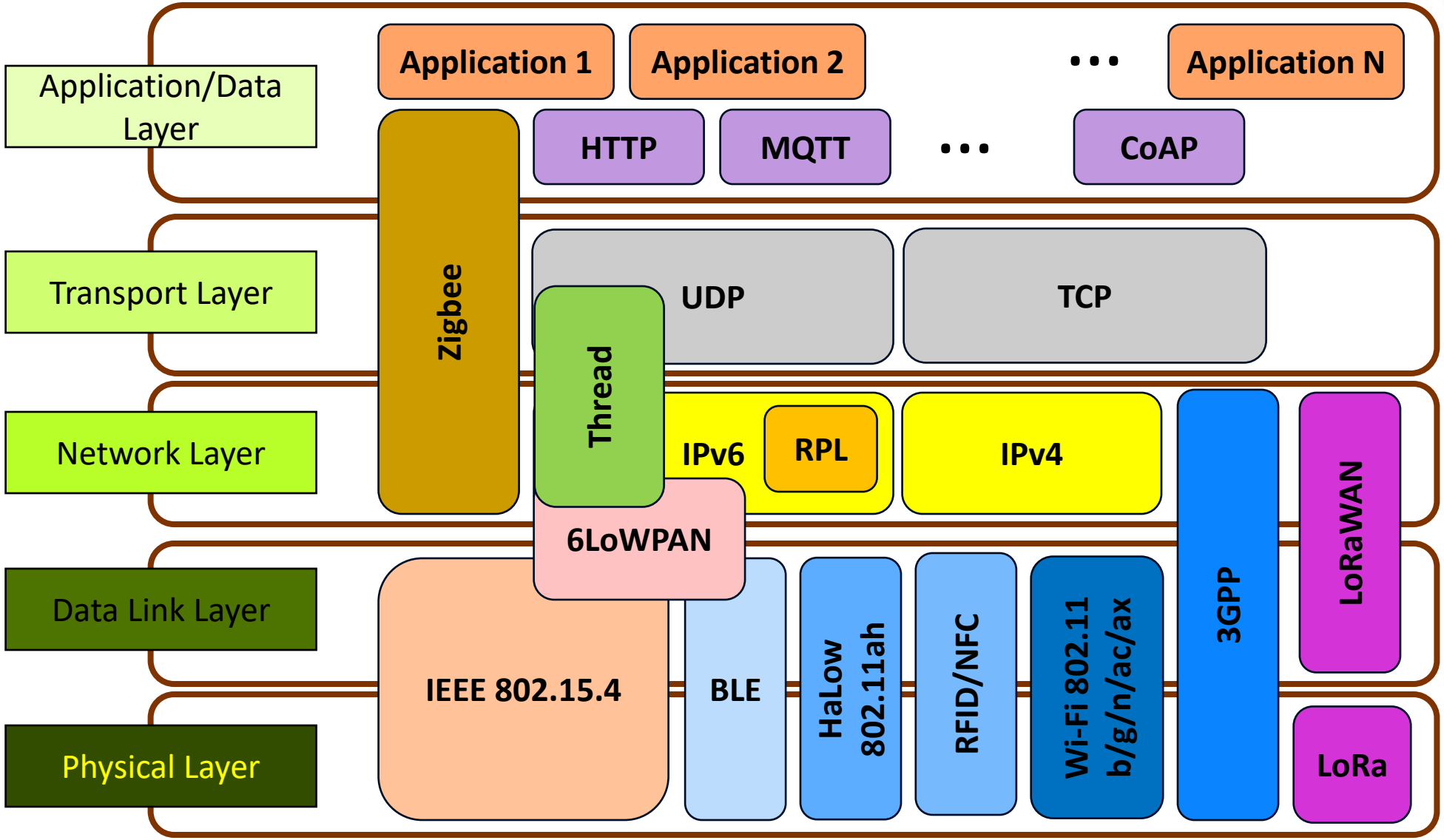


# Communication

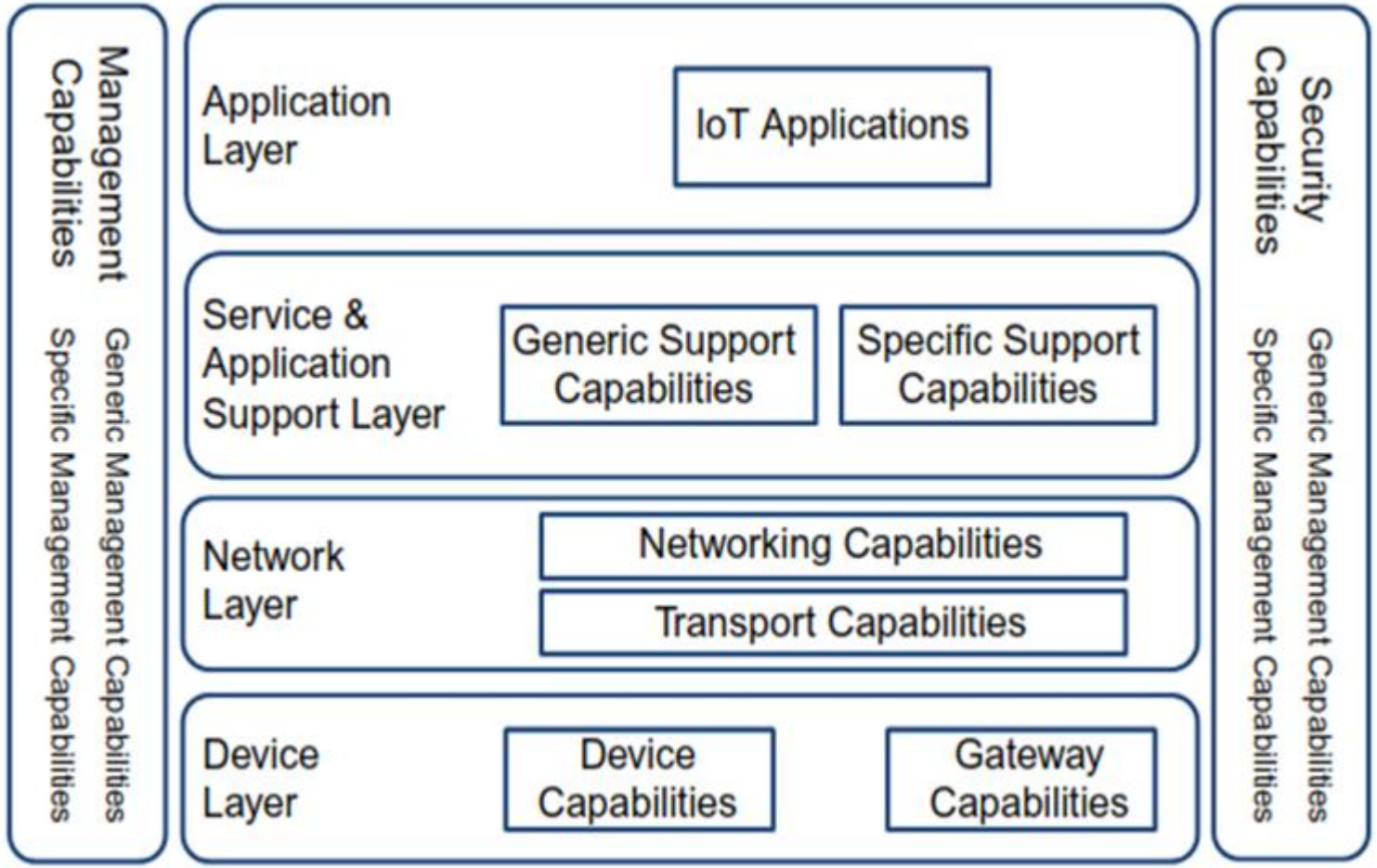
| Technology | Frequency | Data Rate | Range | Power Usage | Cost |
|---|---|---|---|---|---|
| 2G/3G | Cellular Bands | 10 Mbps | Several Miles | High | High |
| Bluetooth/BLE | 2.4Ghz | 1, 2, 3 Mbps | ~300 feet | Low | Low |
| 802.15.4 | subGhz, 2.4GHz | 40, 250 kbps | > 100 square miles | Low | Low |
| LoRa | subGhz | < 50 kbps | 1-3 miles | Low | Medium |
| LTE Cat 0/1 | Cellular Bands | 1-10 Mbps | Several Miles | Medium | High |
| NB-IoT | Cellular Bands | 0.1-1 Mbps | Several Miles | Medium | High |
| SigFox | subGhz | < 1 kbps | Several Miles | Low | Medium |
| Weightless | subGhz | 0.1-24 Mbps | Several Miles | Low | Low |
| Wi-Fi | subGhz, 2.4Ghz, 5Ghz | 0.1-54 Mbps | < 300 feet | Medium | Low |
| WirelessHART | 2.4Ghz | 250 kbps | ~300 feet | Medium | Medium |
| ZigBee | 2.4Ghz | 250 kbps | ~300 feet | Low | Medium |
| Z-Wave | subGhz | 40 kbps | ~100 feet | Low | Medium |

Source: https://blog.helium.com/802-15-4-wireless-for-internet-of-things-developers-1948fc313b2e

# Current Protocols for IoT



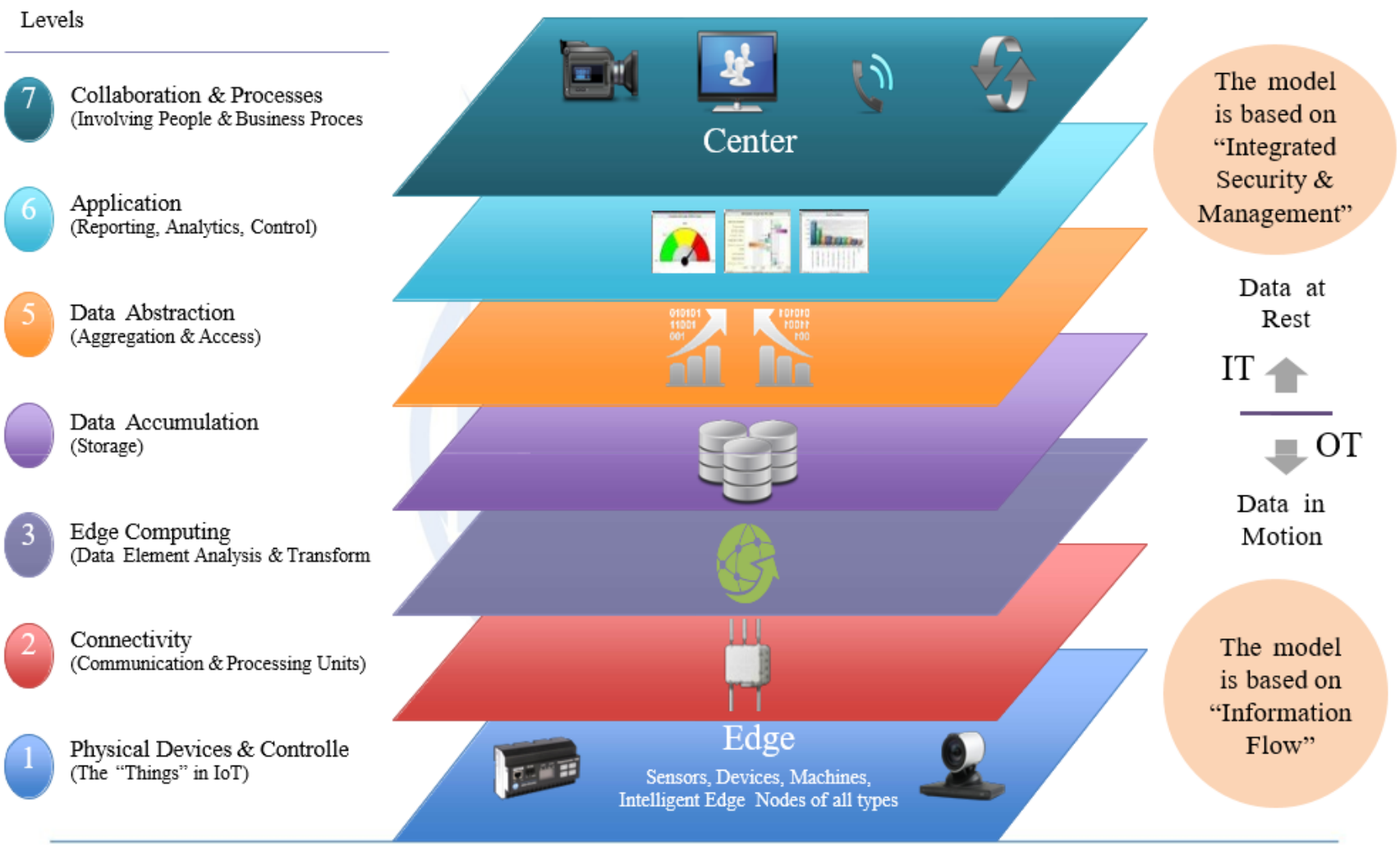
# Layered functional capabilities for IoT systems




IoT Reference Model (ITU-T Recommended)

*https://www.itu.int/rec/T-REC-Y.2060-201206-I*

# Data-centric view of IoT operations



# IoT Design Requirements

| IoT Network | Impact on IoT Systems Design |
|---|---|
| **Scale** | **Tens of thousand sensors in a given site; or millions distributed geographically**<br>More pressure on application architectures, network load, traffic types, security, non-standard usage pattern |
| **Heterogeneous end-points** | **Vast array of sensors, actuators, and smart devices – IP or non-IP**<br>Diverse data rate exchange, form factor, computing and communication capabilities, legacy protocols |
| **Accessibility-Visibility of end-points** | **May be deployed before activation, maybe or cannot-be accessed once deployed**<br>Devices deliver services with little or no human control, difficult to correct mistakes, device management is key |
| **Criticality of services** | **Human life critical (Healthcare), Critical infrastructure (Smart Grid)**<br>Stringent latency (10ms for SG) and reliability requirements, may challenge/exceed network capabilities of today |
| **Intrusiveness** | **Things with explicit intent to better manage end-users (eHealth, Smart Grid)**<br>Issues of Privacy become major obstacles |
| **Geography** | **Movement across borders**<br>Issues of numbering for unique identification |

*Source: ITU CoE training on BB networks planning, Bangkok, Sep 2017*

# IoT Network Connectivity Requirements

| IoT Network | Impact on IoT Systems Design |
|---|---|
| **Resource-constrained endpoints** | **Severely resource constrained (memory, compute)**<br>Cost motivation: compute/memory several orders of magnitude lower, limited remote SW update capability, light protocols, security |
| **Low Power** | **Some end-point types may be mostly 'sleeping' and awakened when required**<br>• Sensors cannot be easily connected to a power source<br>• Reduced interaction time between devices and applications (some regulations state duty cycle of no more than 1%)<br>• Idle mode most of the time (energy consumption of around 100 µW). Connected mode just for transmission (mA)<br>• < 100 MHz clock frequency<br>• Embedded memory of few Mb |
| **Embedded** | **Smart civil infrastructure, building, devices inside human beings**<br>Sensors deployed in secure or hostile operating conditions, difficult to change without impacting system, Security |
| **Longevity** | **Deployed for life typically, have to build-in device redundancy**<br>Very different lifetime expectancy, rate of equipment change in IoT business domains much lower than ICT Industry |
| **High Sensitivity on reception** | **Gateways and end-devices with a high sensitivity around**<br>-150 dBm/-125 dBm with Bluetooth lower than -95 dBm in in cellular |

*Source: ITU CoE training on BB networks planning, Bangkok, Sep 2017*

# Key IoT Network Metrics

- **Latency (Round-Trip Time)**
  - What It Is: Time between sending data and receiving an acknowledgement or echo.
  - Why It Matters: Critical for real-time control or interactive applications.
- **Throughput (Data Rate)**
  - What It Is: Amount of data transferred per unit time (bps, Kbps, Mbps).
  - Why It Matters: Determines how fast you can send sensor readings or firmware updates.
- **Packet Loss**
  - What It Is: Fraction of transmitted packets that never arrive or get acknowledged.
  - Why It Matters: Indicates reliability; high loss can break important IoT workflows.
- **Jitter (Latency Variation)**
  - What It Is: Fluctuation in packet delay over time.
  - Why It Matters: Affects applications sensitive to timing (voice, motion control).

# Obtaining These Metrics (No Dedicated Tools)

- **Custom Send-and-Reply Logic**
  - Write a simple application that sends a known number of packets and measures send/receive timestamps.
  - How to Do It:
    1. Start a timer when you send packet #1.
    2. Stop the timer once you receive all echoes (or acknowledgements).
    3. Record how many packets arrived vs. were sent.
    4. Transmit a large volume of packets to thoroughly stress-test the network and obtain more representative average metric readings.
    5. Calculate:
       - Latency: (ReceiveTime - SendTime) for each packet.
       - Throughput: (Total Bytes * 8) / Elapsed Time.
       - Packet Loss: (Sent - Received) / Sent.
       - Jitter: Variation in consecutive packet latencies.

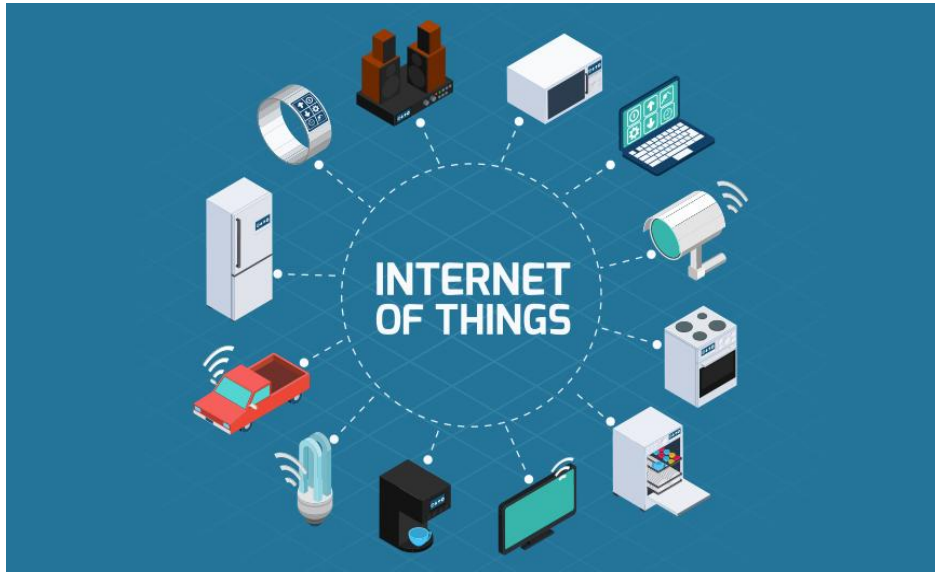
# Obtaining These Metrics (No Dedicated Tools)

- **Adapt to Each Protocol**
  - Wi-Fi: Send UDP/TCP data directly over IP.
  - Bluetooth/BLE: Use a custom GATT characteristic to send data and receive notifications.
  - LoRaWAN/Zigbee: Rely on sending data uplinks and receiving confirmations.

- **Practical Constraints**
  - Collect Logs: Store timestamps in your microcontroller or gateway to compute metrics offline.
  - Duty Cycle & Battery: For LoRa/Zigbee, limit how frequently you send.
  - Payload Size Limits: Some protocols allow only small packets
    - segment large data accordingly.
  - Clock Sync (Optional): To measure one-way delay or advanced jitter, ensure devices have a common time reference (NTP, etc.).

# AIoT: Fusing AI with IoT

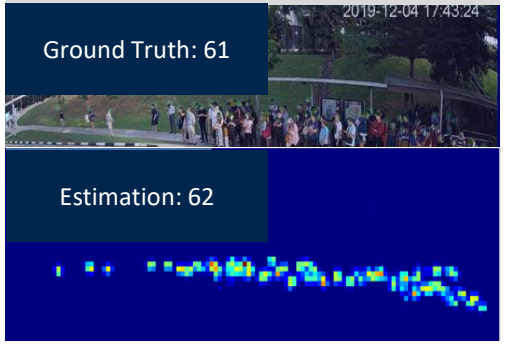
## Traditional IoTs



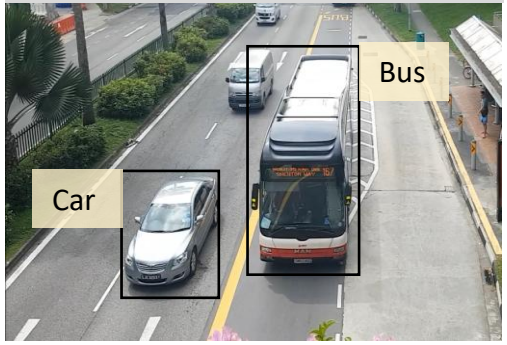

Source: https://www.bankinfosecurity.com/gao-assesses-iot-cybersecurity-other-risks-a-9926

## AIoTs

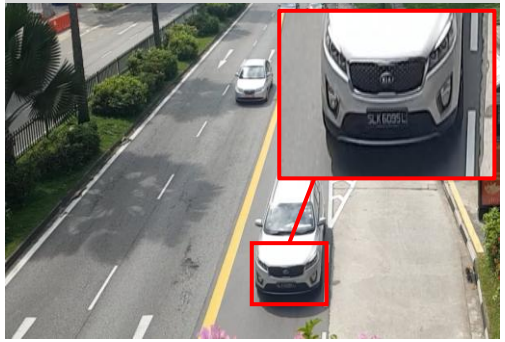
**Crowd Counting**




**Object Localization & Classification**

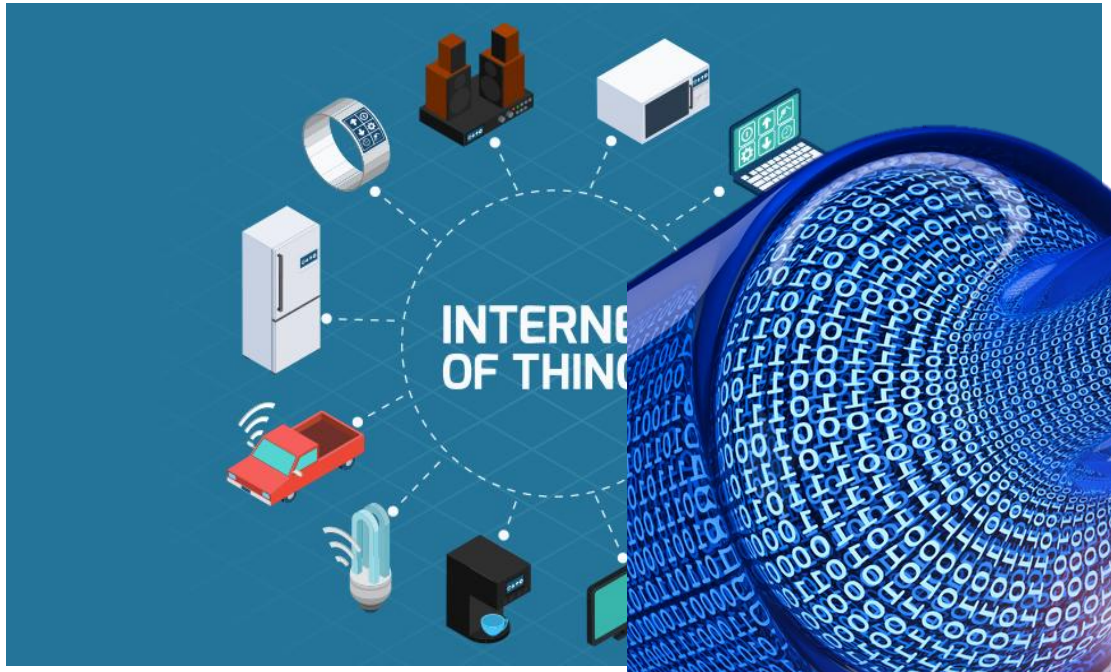



**License Pate Recognition**



# The Need for 'Edge Computing'

- IDC estimates that there will be **41.6 billion connected IoT** devices generating a whopping **79.4 zettabytes of data** in 2025[1].

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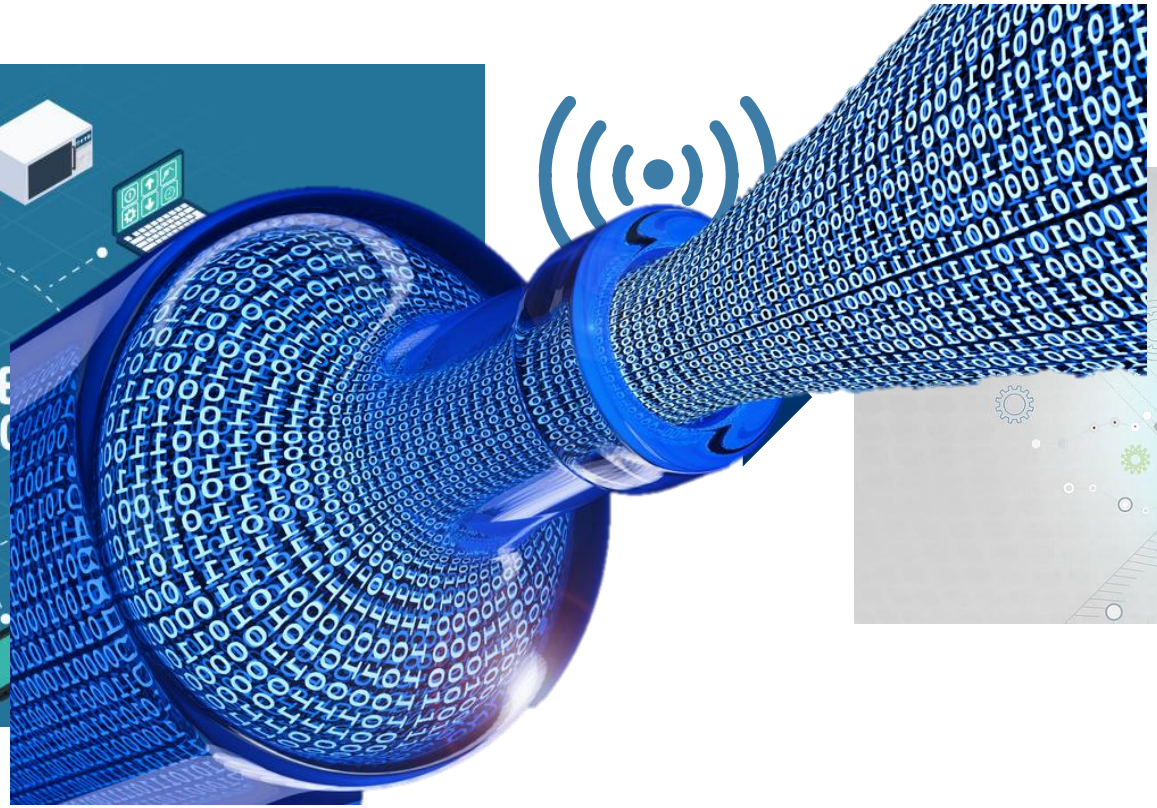
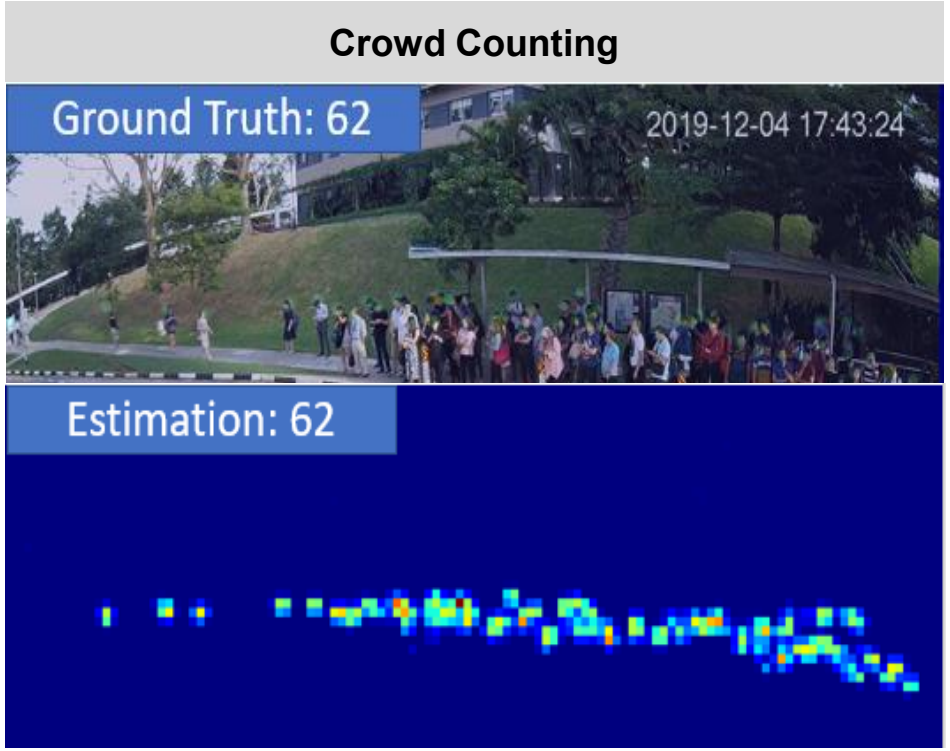


Source: https://www.hpcwire.com/

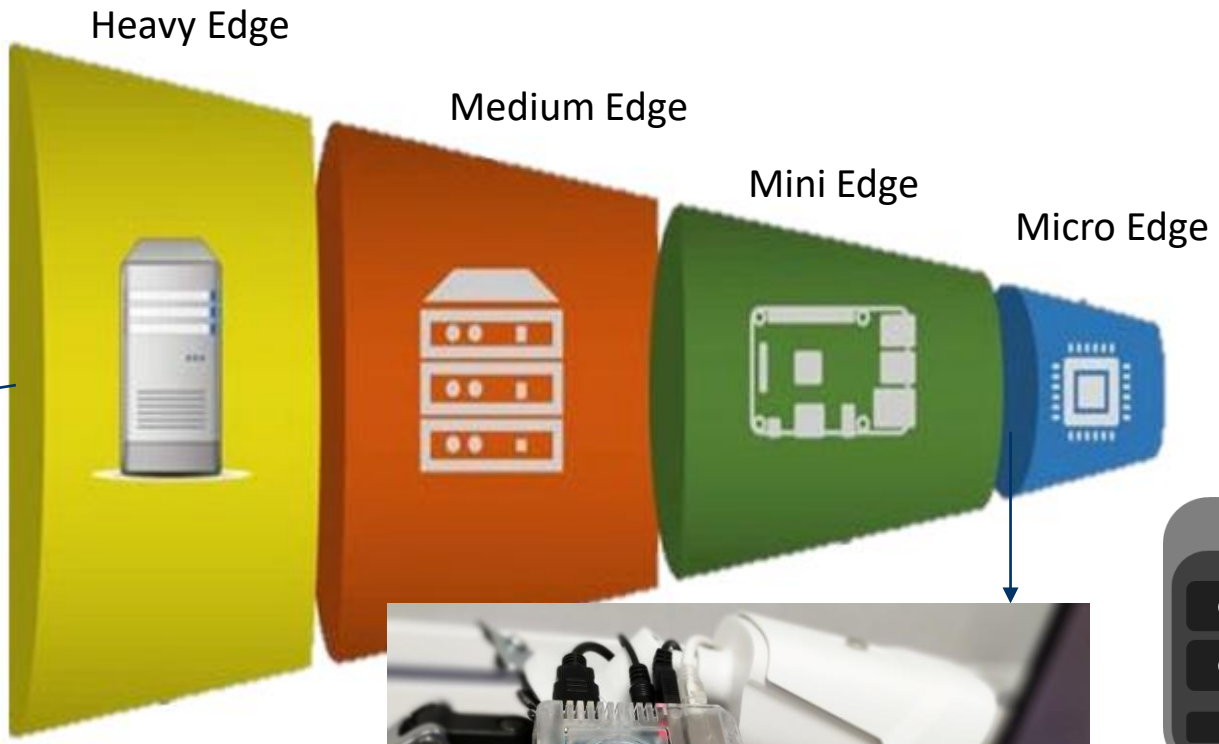
[1] Worldwide Global DataSphere IoT device and data forecast (2019-2023), IDC

# Advantages of Edge Computing

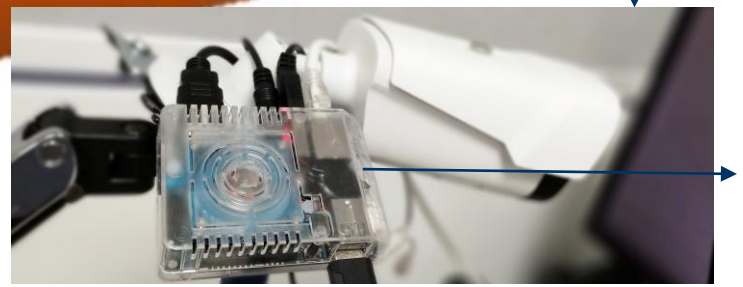
- Reduces Network Bandwidth Requirement
- Faster response time
- Significantly reduced data storage requirements
- …

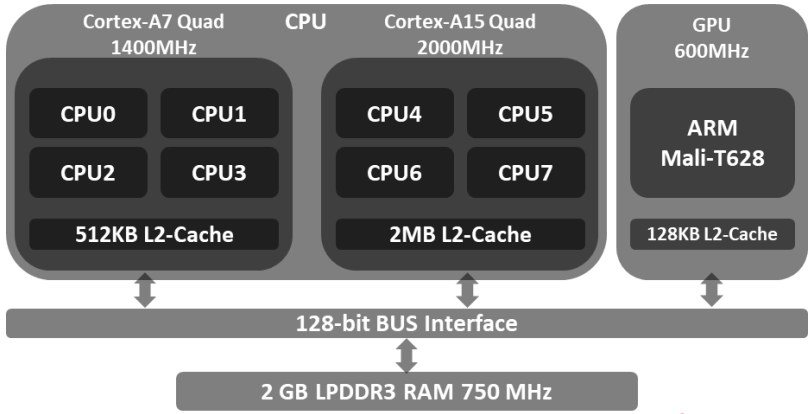


# Edge Computing Platforms: Good old Embedded Systems



Credit:
Janakiram.com






Heterogeneous Edge Computing Platforms

**So how do we map our applications effectively on such heterogeneous platforms while considering power and performance constraints?**

# Designing for the Edge: Key Considerations

1. Latency and Response Time
2. Data Processing and Analysis Capabilities - Locally
3. Scalability - Number of devices and the Geographical
4. Security and Privacy, i.e. privacy concerns
5. Network Connectivity and Bandwidth
6. Hardware Constraints
7. Energy Efficiency
8. Data Management and Storage
9. Robustness and Reliability
10. User Interface and Experience
11. Regulatory Compliance
12. Cost-Effectiveness
13. Customization and Flexibility

# [Summary] Factors to Consider

- **Range**
  - Communication distance affects protocol choice
  - Short-range: Bluetooth, Zigbee
  - Long-range: LoRaWAN, Sigfox
- **Power Consumption**
  - Critical for battery-powered devices
  - Low-power protocols: BLE, Zigbee
- **IoT Network Metrics**
  - Latency (response time)
  - Throughput (data rate)
  - Packet Loss
  - Jitter (latency variation)
- **Compatibility & Scalability**
  - Works with existing devices and infrastructure
  - Supports large-scale deployments
- **Security**
  - Strong encryption & authentication
  - Examples: TLS, DTLS for sensitive data
- **Cost**
  - Hardware & maintenance expenses
  - Specialized protocols may increase cost